

DESIGN PATTERNS LAB

CREATIONAL DESIGN PATTERNS ASSIGNMENT

Online Examination Subsystem

Programme Title	BSSE
Module Title	Design Patterns Laboratory
Assignment Title	Factory Method, Abstract Factory, Builder and Question Sourcing
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BSSE Roll	BSSE1625
Instructor	Mridha Md. Nafis Fuad
Submission Date	27 June 2026
Public Repository	https://github.com/tawratnibir/online-examination-design-patterns
Required PDF Name	BSSE1625.pdf

Submission contains the required UML class diagrams, public source-code link, and public AI-chat evidence link.

SUBMISSION DELIVERABLES

Required deliverable	Location in this document	Status
Solution UML class diagrams	Figures 1-8	Included
Publicly accessible source-code link	Repository section below	Included
Chats made with an AI assistant	AI Assistance Chat Evidence section	Included

Public Source-Code Repository

<https://github.com/tawratnibir/online-examination-design-patterns>

UML Notation Used

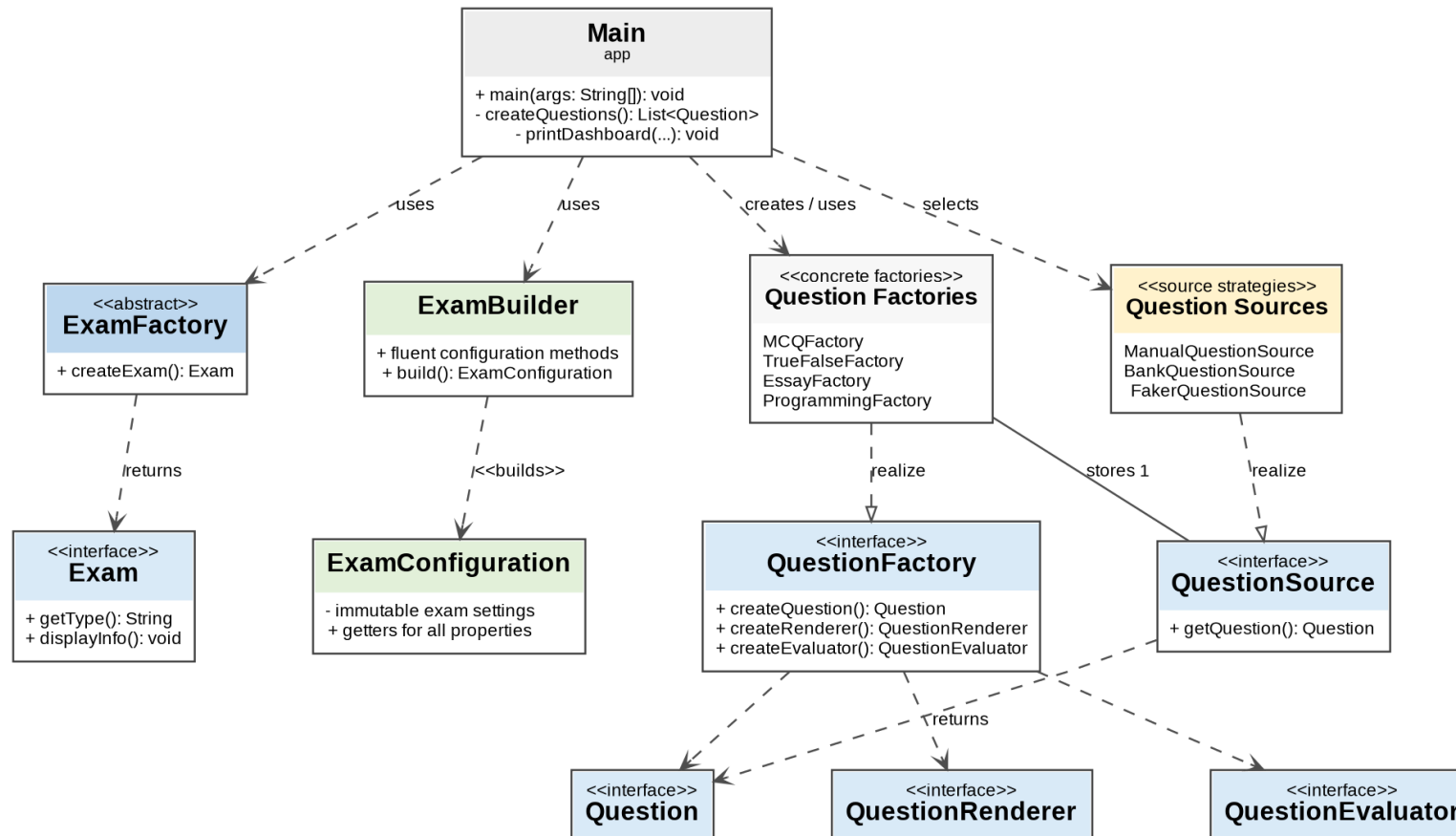
Symbol / relation	Meaning
+ / - / ~	public / private / package visibility
Solid line with hollow triangle	generalization (class inheritance)
Dashed line with hollow triangle	realization (implements an interface)
Dashed arrow labelled <<create>>	direct object-creation dependency
Dashed arrow labelled <<returns>>	object is returned without direct construction
Solid line with multiplicity 1	stored association to exactly one object

Diagram Index

- Figure 1 - Overall Online Examination Subsystem
- Figure 2 - Factory Method Pattern (exam provisioning)
- Figure 3 - Abstract Factory: MCQ family
- Figure 4 - Abstract Factory: True/False family
- Figure 5 - Abstract Factory: Essay family
- Figure 6 - Abstract Factory: Programming family
- Figure 7 - Builder Pattern (exam configuration)
- Figure 8 - Question sourcing and constructor dependency injection

SOLUTION UML CLASS DIAGRAMS

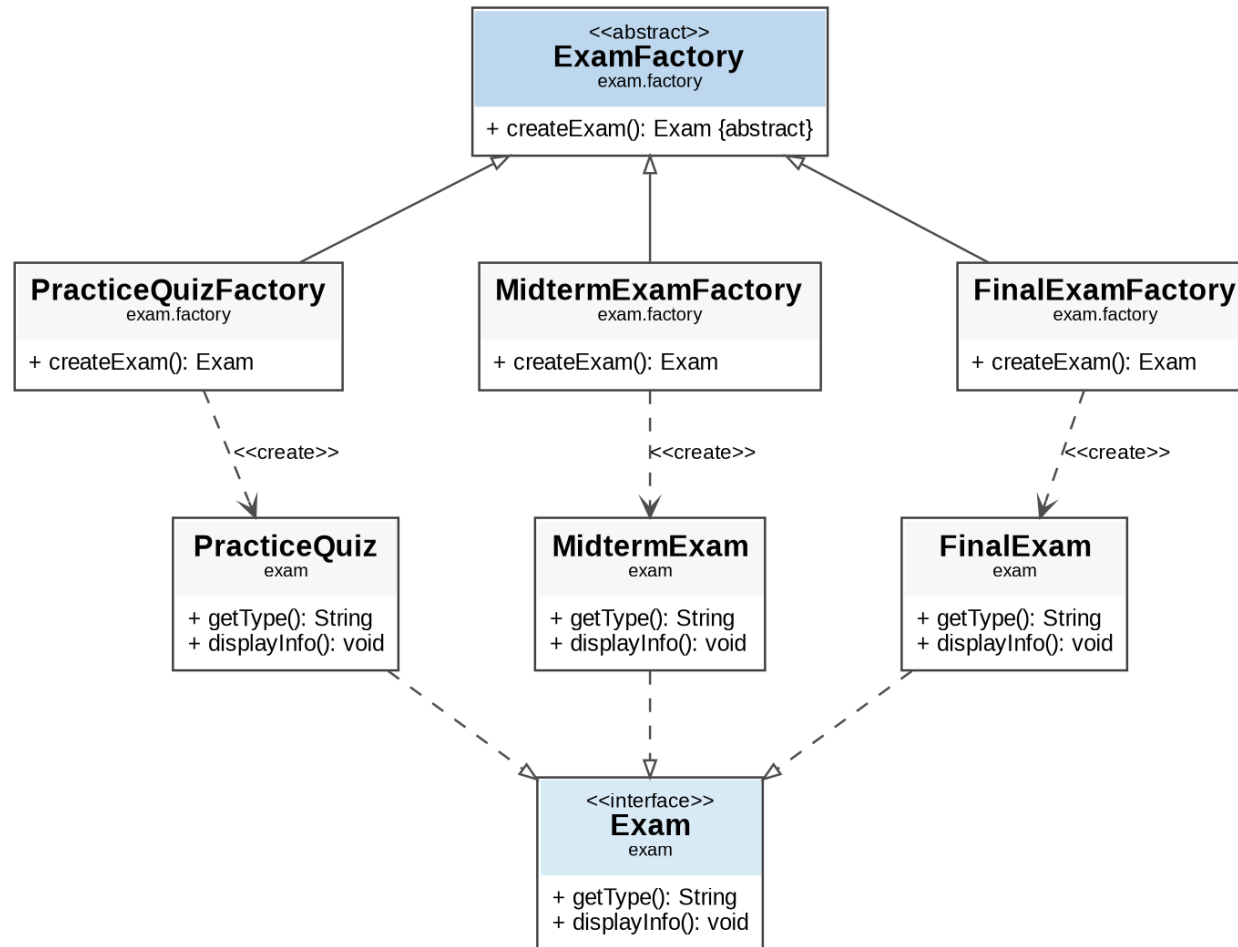
Figure 1. Overall Online Examination Subsystem



This overview connects the client, Factory Method, Builder, Abstract Factory and QuestionSource strategy abstractions used by the final application.

SOLUTION UML CLASS DIAGRAMS

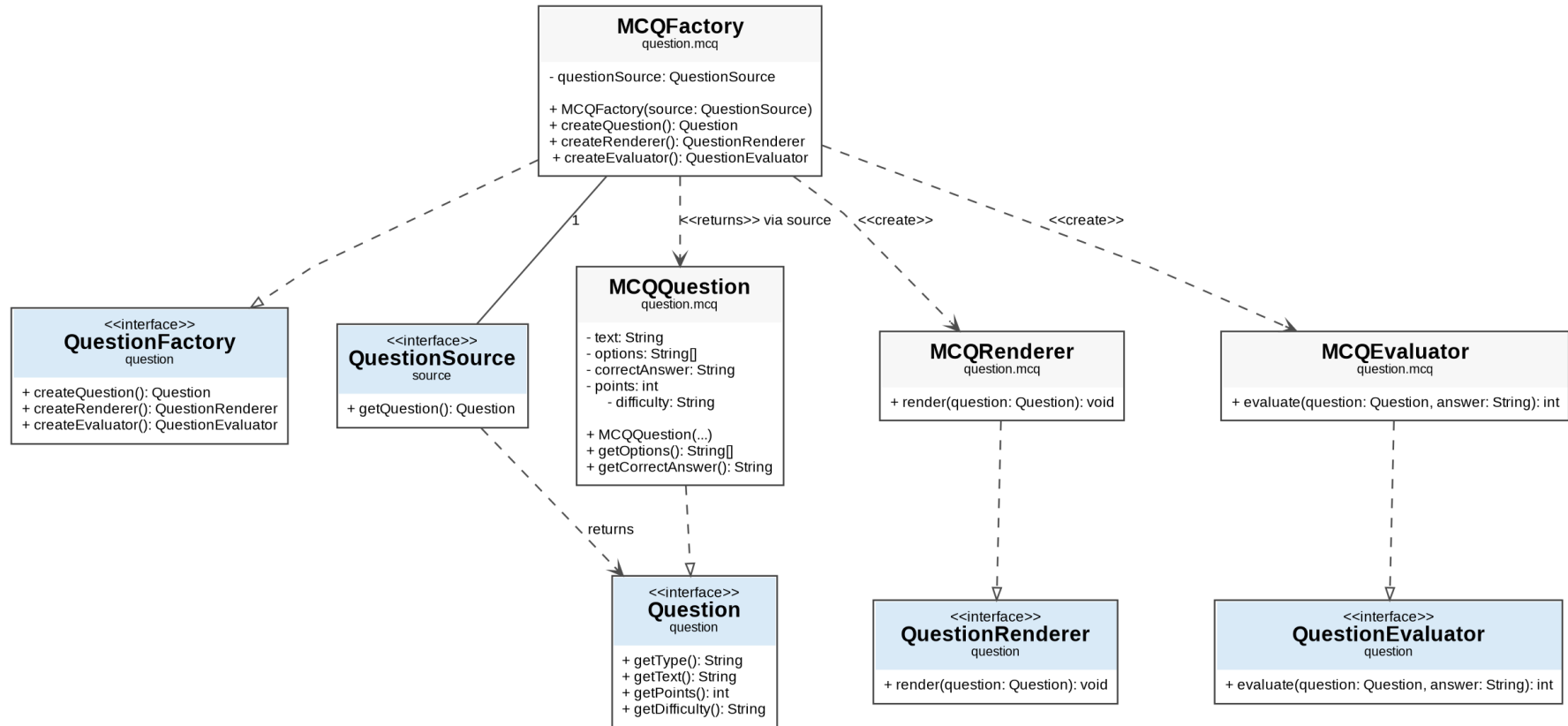
Figure 2. Factory Method Pattern - Exam Provisioning



Concrete exam factories inherit from ExamFactory, while concrete exams realize the Exam interface. Each factory directly creates its corresponding exam product.

SOLUTION UML CLASS DIAGRAMS

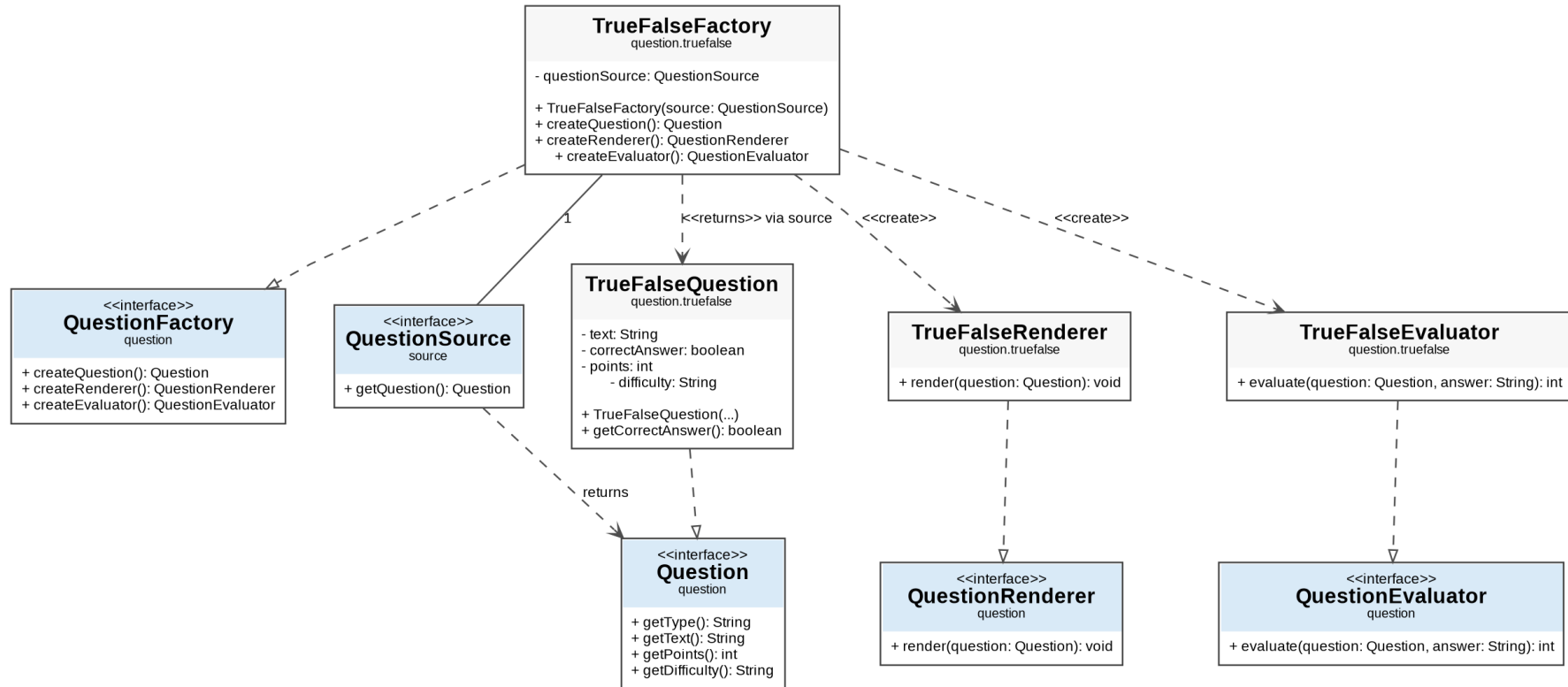
Figure 3. Abstract Factory Pattern - MCQ Product Family



MCQFactory realizes QuestionFactory and stores one QuestionSource. It returns an MCQQuestion obtained through that source, while directly creating the matching MCQRenderer and MCQEvaluator.

SOLUTION UML CLASS DIAGRAMS

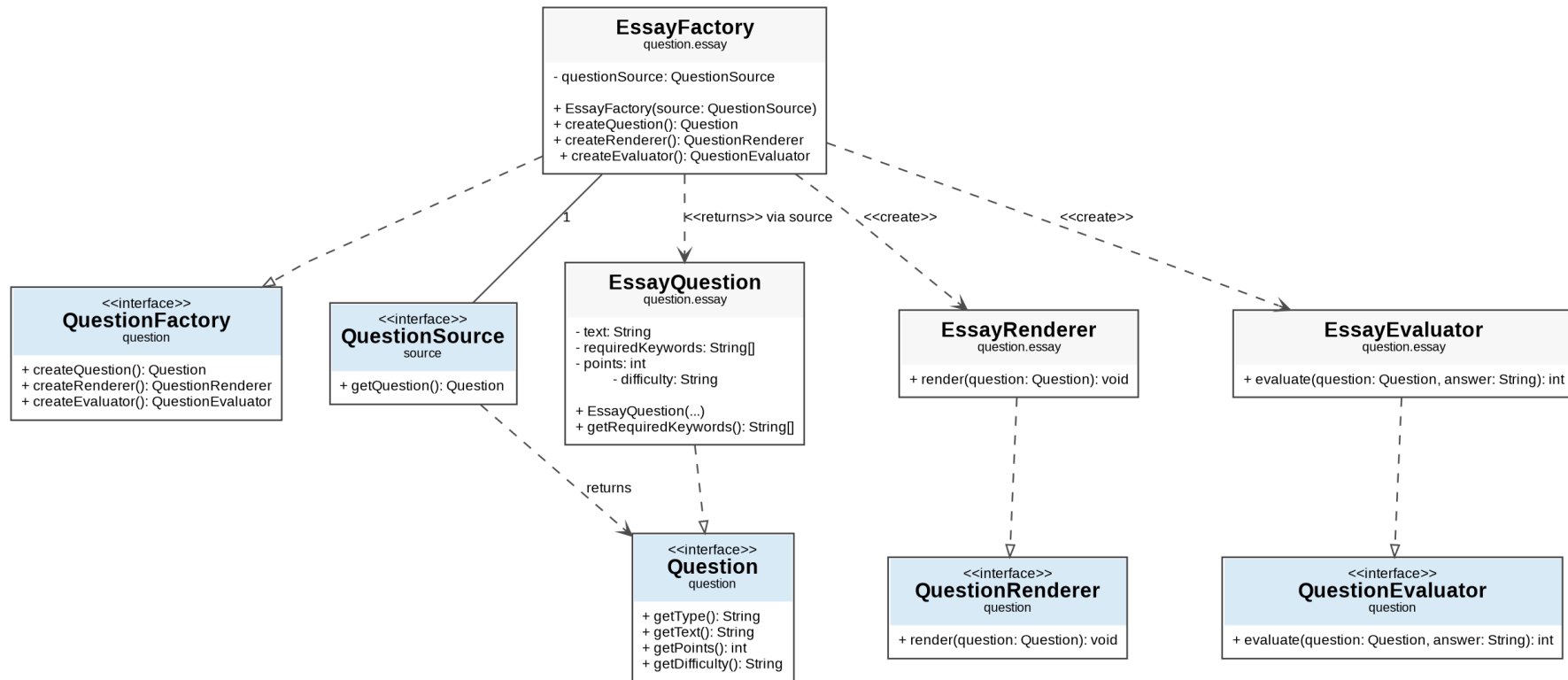
Figure 4. Abstract Factory Pattern - True/False Product Family



TrueFalseFactory returns a source-provided TrueFalseQuestion and directly creates the compatible TrueFalseRenderer and TrueFalseEvaluator products.

SOLUTION UML CLASS DIAGRAMS

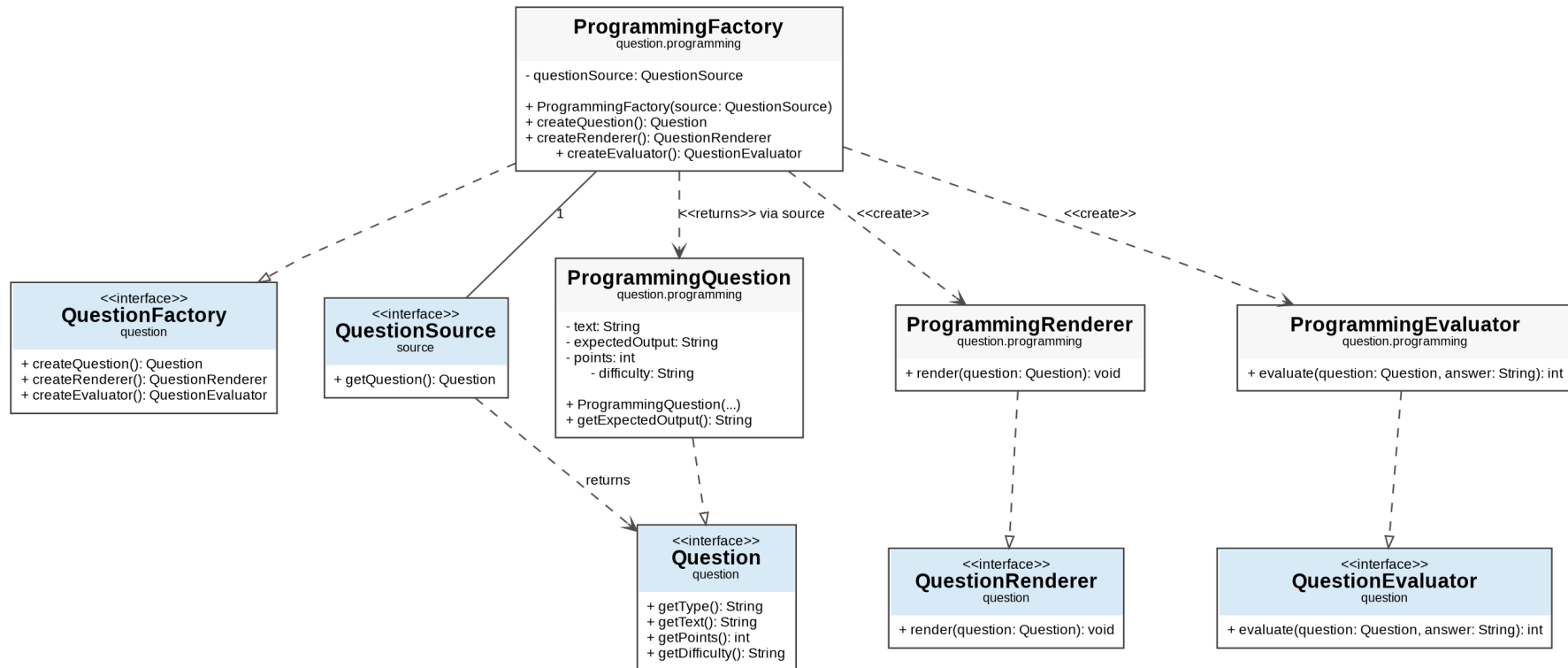
Figure 5. Abstract Factory Pattern - Essay Product Family



EssayFactory preserves family compatibility: the EssayQuestion is obtained through QuestionSource, while EssayRenderer and EssayEvaluator are directly constructed by the factory.

SOLUTION UML CLASS DIAGRAMS

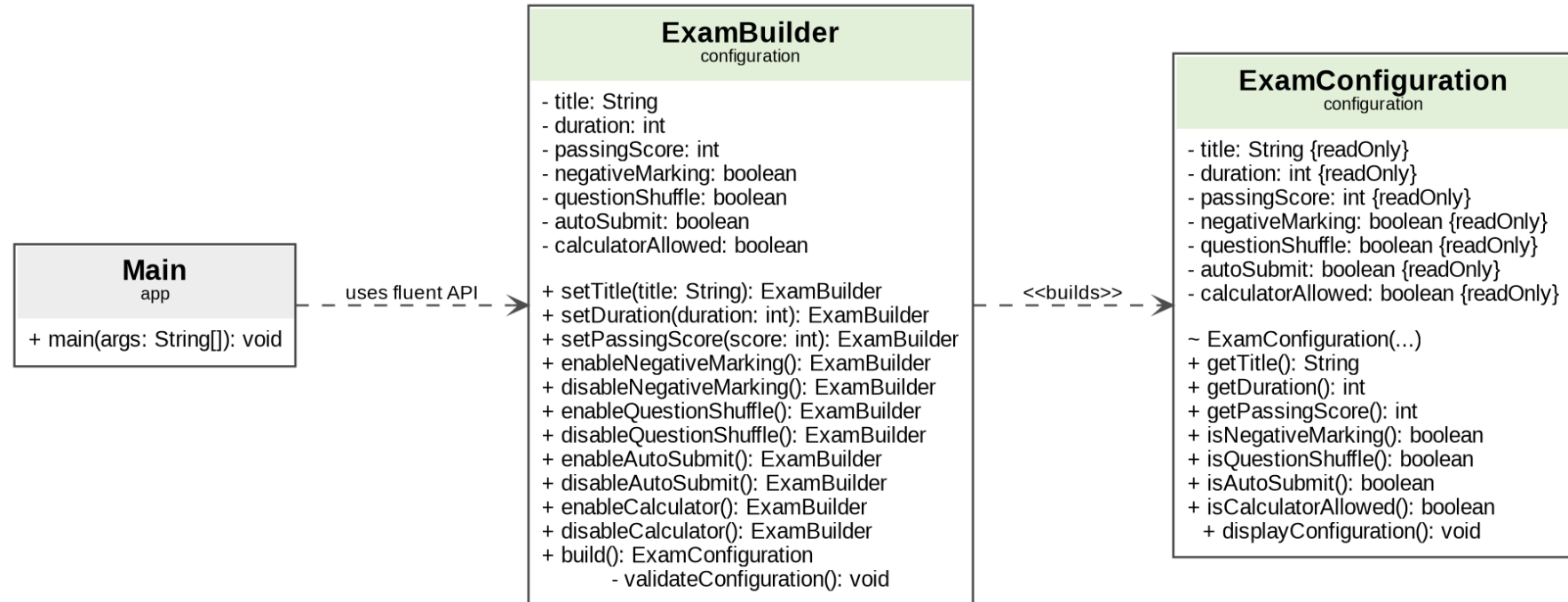
Figure 6. Abstract Factory Pattern - Programming Product Family



ProgrammingFactory returns a ProgrammingQuestion supplied through QuestionSource and directly creates the associated renderer and evaluator.

SOLUTION UML CLASS DIAGRAMS

Figure 7. Builder Pattern - Immutable Exam Configuration



ExamBuilder stores temporary values, exposes the exact fluent methods implemented in the source code, validates the settings, and builds an immutable ExamConfiguration.

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Figure 8(a). Question Source Strategies

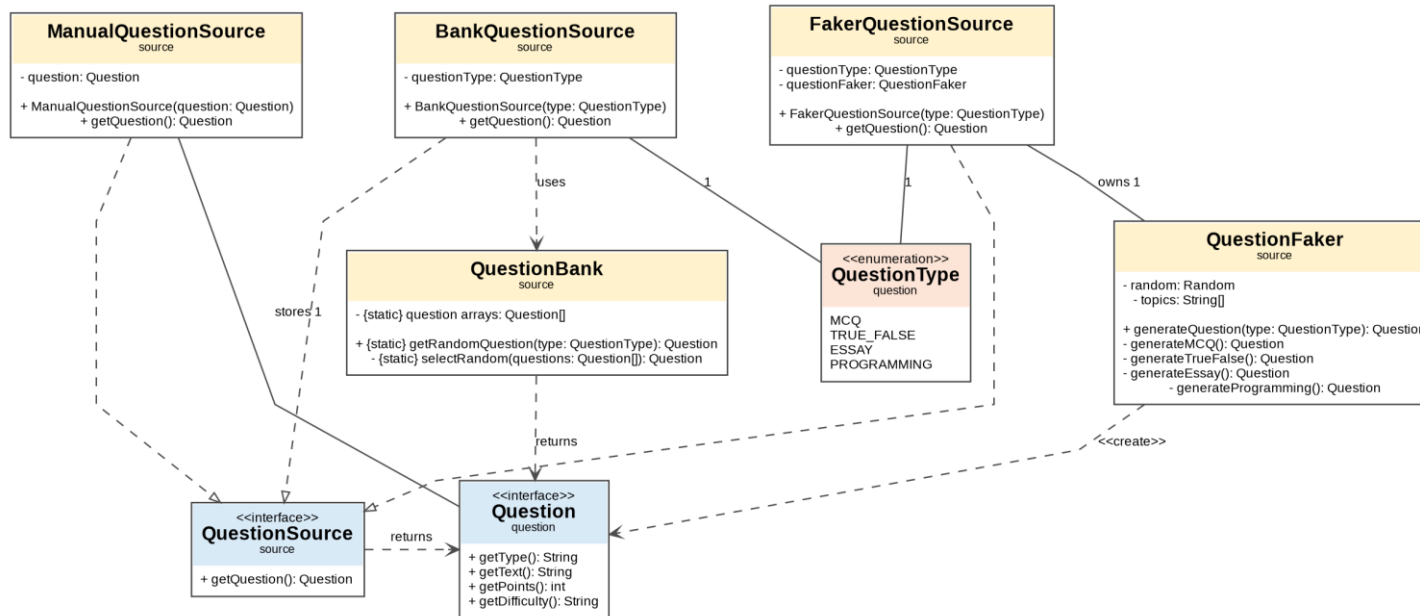
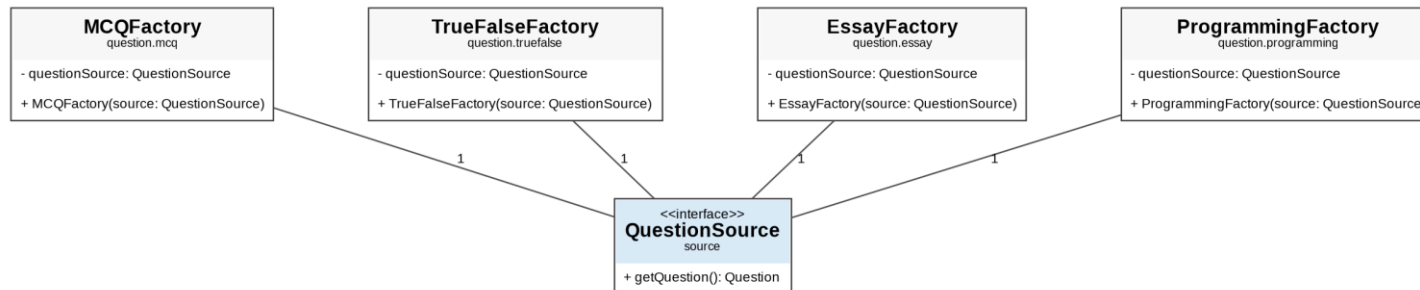


Figure 8(b). Constructor Dependency Injection into Question Factories



AI ASSISTANCE CHAT EVIDENCE

The complete AI-assisted discussion used during the development of this assignment is available through the public ChatGPT share link below.

Chat evidence topic	Coverage in Shared Chat
1. Factory Method	Theory discussion and Part A implementation guidance
2. Abstract Factory	Theory discussion and Part B implementation guidance
3. Builder Pattern	Theory discussion and Part C implementation guidance
4. Question Sourcing / IoC	Part D design, enum explanation and dependency injection
5. Integration and UML	Package restructuring, fixes, repository review and document preparation

Public AI Chat Link

<https://chatgpt.com/share/6a400ece-b5f0-83ee-91bb-9f941c1d679e>